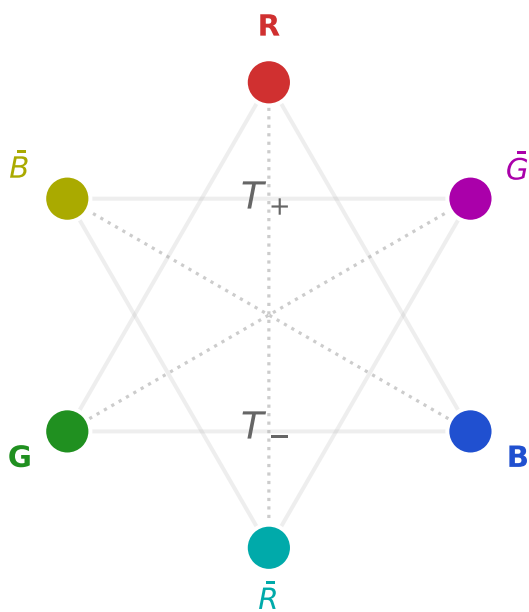
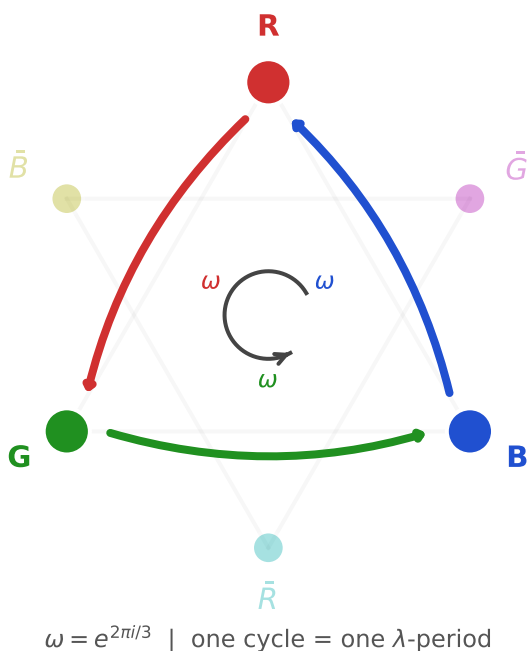


Stella Diagrams: Interactions on $\partial\mathcal{S}$

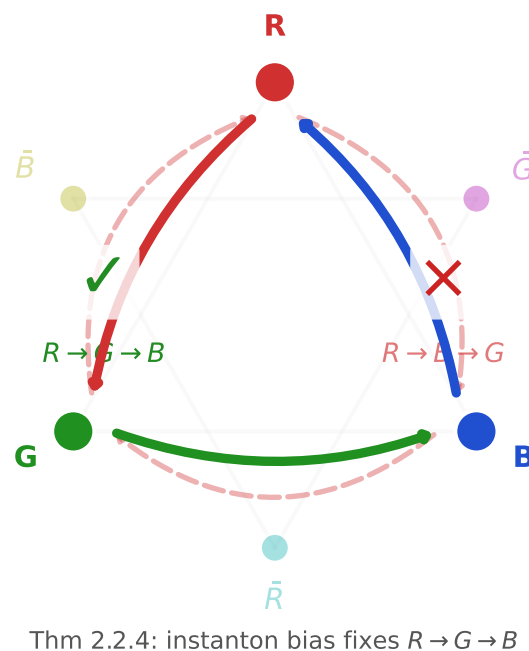
(a) Stella graph $\partial\mathcal{S}$



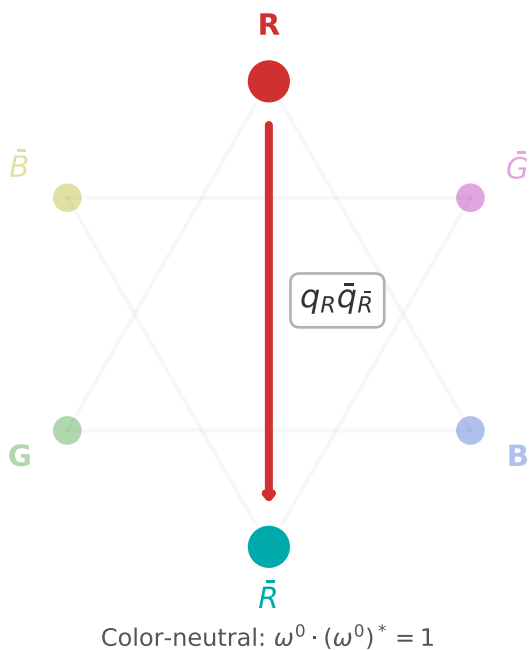
(b) Color cycle $R \rightarrow G \rightarrow B \rightarrow R$



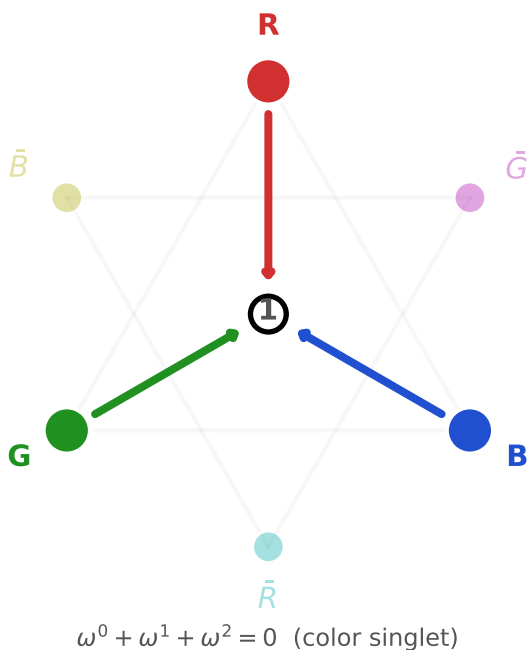
(c) Chirality: $\langle Q \rangle > 0$ selects direction



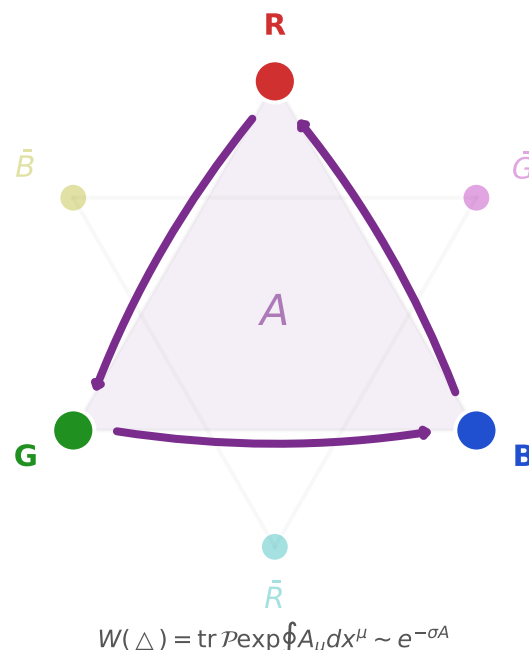
(d) Meson: $q_c \bar{q}_c$



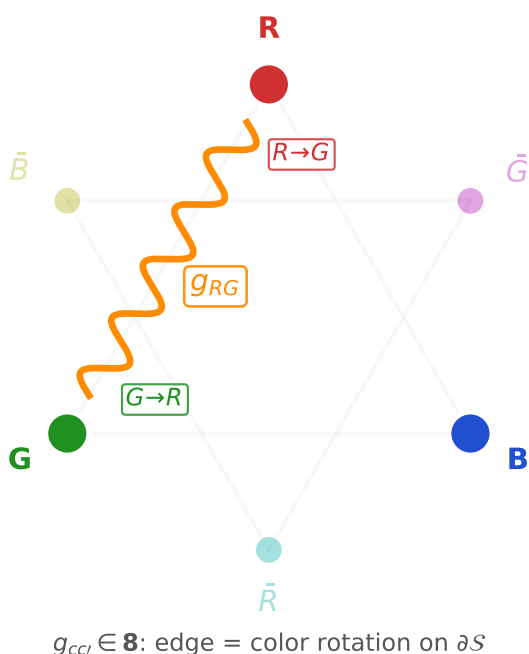
(e) Baryon: $R + G + B \rightarrow \mathbf{1}$



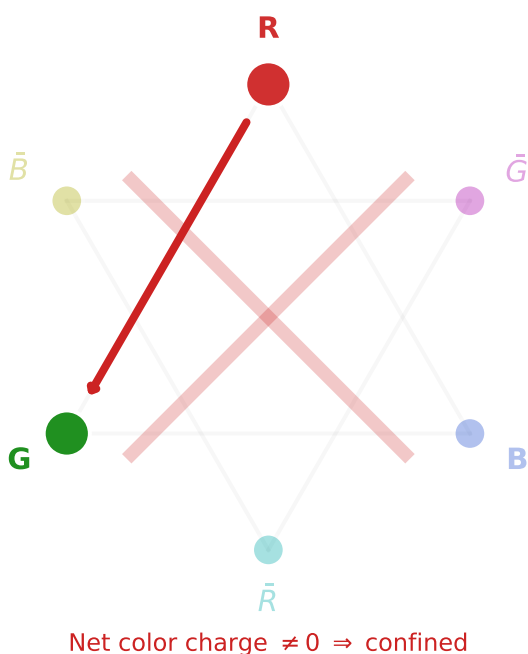
(f) Wilson loop on T_+ face



(g) Gluon: adjoint $\mathbf{8}$ exchange



(h) Forbidden: open $R \rightarrow G$



(i) Stella diagram rules

Element	Symbol	Rule / Physics
Color vertex	$\bullet_c$	$c \in \{R, G, B, \bar{R}, \bar{G}, \bar{B}\}$
Directed edge	$c_1 \rightarrow c_2$	Phase factor $\omega^{\Delta c}$
Wavy line	$\sim \sim$	Gluon ($\mathbf{8}$): color rotation
Closed loop	$\mathcal{U}$	Color-neutral (physical)
Open path	$c_1 \rightarrow c_2$	Confined (unphysical)
Chirality	$R \rightarrow G \rightarrow B$	Fixed by $\langle Q \rangle > 0$
Conjugation	$c \leftrightarrow \bar{c}$	$T_+ \leftrightarrow T_-$ (reflection)
Wilson loop	$\triangle$ on face	Area law $W \sim e^{-\sigma A}$
Face area	A	$\propto R_{\text{Stella}}^2$